

What re-encoding **CHANGES**

the same operator $a_i^\dagger a_j$ becomes a longer or shorter Pauli string

simulation cost

Jordan-Wigner  weight = n

tree encoding  weight $\sim \log n$



**GAUGE-
COVARIANT**

What re-encoding **CANNOT** change

the dynamical Lie algebra and the magic do not change

trainability (1/variance)

dim $\mathfrak{g}$

JW

tree

magic

JW

tree

=



**GAUGE-
INVARIANT**